Why Does Imported Talent Hold the Central Roles in China, but Not in Japan or Korea?

Abstract

Imported talent is widely expected to benefit the local participants around it. Yet research on stars and high performers suggests it can instead command attention and resources, narrowing local players' roles. We explore what role imported talent plays, and how that role varies across leagues. Three event-level dimensions from the Chinese Super League (CSL), Japan's J1, and Korea's K1 (2015–2025) anchor our analysis: 185,021 domestic player-match observations, 51,700 substitution decisions, and 3,255 goal-creating chains. Across these, in-match performance changes little when foreign teammates are absent through injury. What shifts is not the level of play but its role allocation, a foreign-centered architecture: CSL managers substitute foreign-for-foreign rather than a domestic forward, and the lineup routes in-play coordination through foreign finishers. Neither J1 nor K1 reproduces the architecture, even though K1 concentrates foreign players in similarly narrow attacking regions. We link this break to two conditions that coincide in CSL: a foreign-to-domestic talent gap that reaches an order of magnitude, unlike in either neighbor, and acute institutional pressure to exploit imported talent rather than explore. Within CSL, the architecture intensifies as the talent gap widens across team-seasons. This displacement from central roles parallels crowd-out in other high-skilled talent-import settings.

INTRODUCTION

Across many organizational contexts, importing foreign expertise is expected to benefit not only the newcomers but also the local participants around them, raising the performance of the whole. Peer-effects research documents contemporaneous positive productivity spillovers from high-performing colleagues to their coworkers (Mas & Moretti, 2009), and treatments of human capital as a unit-level resource recast such gains as collective capacity emerging through interaction and interdependence (Ployhart, Nyberg, Reilly, & Maltarich, 2014). This expectation underwrites hiring practice, talent-attraction policy, and organizational design alike.

Yet a growing literature on stars and high performers suggests this expectation is conditional. Stars—employees with "disproportionately high and prolonged" performance, visibility, and relevant social capital (Call, Nyberg, & Thatcher, 2015: 624)—can command organizational attention and resources, leaving fewer opportunities for nonstars to lead (Kehoe & Tzabbar, 2015). Central stars in workgroups can foster dependence that dampens peers' own contributions rather than drawing them out (Li, Li, Li, & Li, 2020). The aggregate impact of stars can be positive ("red giants"), negative ("black holes"), or contingent on multilevel conditions (Asgari, Hunt, Lerner, Townsend, Hayward, & Kiefer, 2021). High performers can elicit competitive rather than collaborative responses from peers, with consequences that depend on local conditions (Hendricks, Call, & Campbell, 2023); at the dyadic level, peer responses combine support and undermining as peers protect their share of finite team resources (Campbell, Liao, Chuang, Zhou, & Dong, 2017). More fundamentally, organizations that lean on imported expertise may favor the exploitation of that expertise over exploration, at the expense of their own longer-term adaptiveness (March, 1991). Under such conditions, imports may reshape the local role structure, narrowing rather than broadening the roles local players come to occupy. We explore what role imported talent plays, and under what conditions this role changes.

Professional football (soccer) offers a setting in which this question can be examined directly and at scale: who occupies the central roles, and how the ball moves among them, are recorded play by play. Recent management-science studies of European football suggest that the value of talent there turns on how it is arranged, not its quantity alone. Loignon, Fonti, Bagherzadeh, Gurca, and Shrestha (2025) found an inverted-U relationship between team talent and performance in top-tier leagues, with within-team social capital configurations shaping the curve. Glennon, Morales, Carnahan, and Hernández (2025) showed that immigrant players raise European club performance not only through individual talent but also through a coordinating role that helps clubs deploy a wider range of tactics.

The Chinese Super League (CSL) imported foreign players at a scale then unmatched outside Europe. Over 2011–2025, CSL clubs spent billions of euros recruiting Premier League winners, Champions League veterans, and World Cup squad members, branding the effort "money football." Japan's J1 League (J1) and Korea's K League 1 (K1) ran foreign-player quotas of similar structure at a fraction of that outlay. Across China, Japan, and Korea, we examine whether such spending bought the broadening role that imported talent is expected to play.

These three leagues form an unusually well-matched comparison. China, Japan, and Korea field East Asian first divisions that professionalized within roughly a decade of one another (Korea 1983, Japan 1993, China 1994), under the same confederation, contesting the same continental tournament, on a shared regional template of single-table, promotion-and-relegation play. All three cap foreign participation through quotas of the same broad family, converging toward roughly five foreign players on the pitch. Because the sport, its rules, and the broader institutional framework are shared, a role structure that surfaces in one league but not the others cannot be a property of football itself, or of the roster regulations common to all three. What the leagues do not share is the foreign–domestic talent gap that CSL's spending opened. Corroborating this reading, the role structure intensifies, within CSL, as that gap widens.

The spending bought concentration, not breadth: a *foreign-centered architecture*, in which foreign players hold the central offensive roles while domestic players are positioned around them as peripheral support. It surfaces simultaneously in two distinct places: in how managers substitute, and in who the goal-creating play runs through. This is a matter of role allocation, not output: domestic players' performance changes little when foreign teammates are absent through injury, so what the architecture redistributes is position and routing, not the level of play. Why it consolidates in the CSL and not in otherwise-similar leagues is the question we take up in the Discussion. The sharpest clue is Korea, which crowds its foreign players into the same narrow attacking zone as China, yet they never come to dominate the play as China's do; the same space is put to a different use.

Our exploration contributes to research on the conditions that shape the role imported talent plays. That role is not fixed: under broadly similar quota regimes, CSL builds around its foreign players the architecture that neither J1 nor K1 reproduces. What sustains it in CSL but not in its neighbors remains genuinely uncertain. Drawing on the literatures on high-performer peer effects, human capital resources, and organizational learning, we develop two conditions that, where they coincide, may produce it: a wide foreign–domestic talent gap and institutional pressure that rewards leaning on the foreign core now over slower exploration. A third strand, the durability of foreign–domestic working partnerships, helps the architecture persist within CSL but does not travel across leagues.

THE PHENOMENON

In December 2016, Carlos Tevez left Argentina's Boca Juniors for Shanghai Shenhua of the Chinese Super League, reportedly becoming the highest-paid player in world football (Sky Sports, 2016). He was not alone. From 2011 to 2025, the CSL mounted the most expensive foreign-talent import wave then attempted outside Europe: Anelka and Drogba in 2012, Hulk for €55.8 million in 2016, and Oscar for £60 million in 2017, on terms Europe's top-five clubs would not match (Panja, Chang, & Browning, 2016; RTÉ Sport, 2016; Twomey, 2016). Much of this spending was underwritten by state-backed clubs; at Guangzhou Evergrande, it bought two Asian Football Confederation (AFC) Champions League titles, in 2013 and 2015.

As the spending outran any precedent, regulators kept rewriting the rules that governed foreign and local participation: a 100% tax on high-priced foreign transfers in 2017, a €3 million cap on foreign-player wages in 2020, and recurring revisions to foreign-player quotas and under-23 (U23) requirements. What this churn kept circling was not the scoreboard but a question of composition: how much of the game foreign players should occupy, and how much to leave to the domestic players around them.

That question is not China's alone. Japan's J1 and Korea's K1 recruit foreign players under broadly comparable rules. If the split between imported and domestic players were simply a function of the import rules, all three leagues would settle on much the same division; whether they do is what this paper sets out to discover.

METHODS

Sample

Our match-level dataset draws primarily from Sofascore (https://www.sofascore.com), a sports research firm that publishes live in-match event data. From Sofascore we obtain three categories of data: (1) per-player match-level performance statistics used in our performance regressions; (2) match-level substitution data used in our manager-level mechanism analysis; and (3) goal-creating chain sequences and per-player on-ball activity heatmaps used in our chain-composition and spatial analyses. Transfermarkt (https://www.transfermarkt.com) supplies two additional player attributes: time-stamped market values in euros and injury absence-day records.

The data cover all matches in the Chinese Super League (2015–2025 seasons), Japan's J1 League (2017–2025), and Korea's K League 1 (2020–2025). The unit of observation is the player-match: one player's appearance in one match. The full panel contains 254,099 player-matches across the three leagues. Of these, 226,634 are outfield player-matches (CSL: 85,684; J1: 96,849; K1: 44,101), comprising 185,021 domestic player-matches (the analysis universe for our performance regressions) and 41,613 foreign player-matches. Goalkeepers are excluded throughout. Players are classified as foreign or domestic following each host league's regulatory definition.[1] From Sofascore we also collected senior national-team matches over the same period, yielding 5,820 appearances by *dual-context* domestic players, those observed in both their league and their national team.

Measures

Performance outcomes. For each domestic outfield player-match with positive playing time, we use seven outcomes: six adapted from Glennon et al. (2025) and a composite performance rating from Sofascore. These outcomes fall into three classes: skill indicators that should reflect ability (Pass success, Dribble success, Scores), placebo-class measures of involvement, discipline, and

[1] We use *imported talent* for the theoretical construct (non-domestic players brought into a host league through cross-border transfer) and *foreign players* for the regulatory label used in league quota rules. The two terms denote the same population; we use whichever the context calls for. Separately, *cross-country* refers to comparison across our three settings, CSL, J1, and K1.

rotation that should not (Touches, Any yellow, Substituted), and the composite rating.[2] *Pass success* (passing accuracy) is the percentage of passes that are successful (successful passes / total passes), conditional on attempting at least one pass; *Dribble success* is the analogous rate for dribbles. *Touches* is the number of times the player touched the ball during the match. *Scores*, *Any yellow*, and *Substituted* each equal 100 if the player scored one or more goals, received any yellow card, or was withdrawn before the final whistle, respectively, and 0 otherwise; the 0/100 scaling puts effects in percentage points, on the same scale as the success rates. *Rating* is the Sofascore composite.

Foreign-injury treatment. For the player-match performance analysis, player-injury records document the start and end date of each absence, allowing us to identify the foreign players unavailable to each team on each match date. We measure the treatment in two ways. The value-weighted version, *Net Foreign Injury Loss: Value*, sums $\log(1 + \mathrm{MV}_i/1\mathrm{M})$ across the own team's injured foreign players i at the match date and subtracts the analogous quantity for the opponent team, where MV_i is player i's Transfermarkt market value as of the match date, in euros. The count-based version, *Net Foreign Injury Loss: Count*, uses headcount in place of market value. In both cases, positive values indicate that the player's own team has lost more foreign-talent value (or headcount) to injury than the opponent on that match date.

National-team treatment. For the cross-context analysis, the treatment is *National-team match*, an indicator equal to one for senior national-team matches and zero for matches in the player's domestic top-flight league.

Substitution-decision outcomes. For the substitution analysis, the unit of observation is one in-game substitution event (pooled *N* = 51,700 across the three leagues). We use three outcomes,

[2] Sofascore ratings (range 3.0–10.0, baseline 6.5) aggregate a player's on-ball actions adjusted for opposition strength and match context.

all in percentage points (pp): *Foreign replacement* equals 100 if the incoming replacement is foreign; *Domestic forward replacement* equals 100 if the incoming replacement is a domestic forward; *Defensive position shift* equals 100 if the incoming player's position is more defensive than the outgoing player's (goalkeeper = 0, defender = 1, midfielder = 2, forward = 3). The key regressor is *Withdrawn player is foreign*, an indicator equal to 1 if the outgoing player is foreign.

Controls. Performance regressions include player age and age squared, a home-team indicator, and the opponent's pre-match Elo rating.[3] Substitution-decision regressions include the match minute of the substitution, an injury indicator, running goal differential, and own-team foreign-starter count.

Table 1 reports per-match means of these outcomes by country and foreign/domestic status.

Insert Table 1 about here

Empirical Strategy

We bring three event-level dimensions to the puzzle, each at the most natural level of observation: player-match performance (Table 2), manager substitution decisions (Table 4), and the composition of goal-creating chains (Figure 2). A within-player cross-context contrast (Table 3) supplements the first dimension by holding individual capability fixed by construction.

Performance regression (Table 2). Because injury timing is plausibly exogenous to opponent

[3] Opponent Elo is computed in-house using the football Elo update formula as presented in Hvattum and Arntzen (2010), with parameter conventions following the World Football Elo Ratings project (`https://www.eloratings.net/about`), and teams initialized at the conventional 1500 baseline. For the cross-context analysis, opponent Elo is the team's club Elo for league matches and the country's international Elo for national-team matches.

identity as well as to manager selection, we exploit this variation to ask whether (and along which dimension) the loss of foreign teammates moves domestic-player performance. We enter the two injury-loss measures jointly: the value-weighted loss $V_{j,m,t}$ (*Net Foreign Injury Loss: Value*) asks whether the *quality* of absent foreign players matters (a marginal star is worth more than a marginal squad player), and the count-based loss $C_{j,m,t}$ (*Net Foreign Injury Loss: Count*) asks whether *headcount* matters holding value constant. For each of the seven outcomes, we fit:

$$Y_{p,m,t} = \beta_V \, V_{j,m,t} + \beta_C \, C_{j,m,t} + X'_{p,m,t}\gamma + \alpha_p + \delta_{j,t} + \epsilon_{p,m,t} \tag{1}$$

In equation (1), p, m, t, and j index player, match, season, and own team; player fixed effects α_p absorb time-invariant individual ability, and own-team-by-season fixed effects $\delta_{j,t}$ absorb factors constant within a team-season, such as the coaching regime and roster; $X'_{p,m,t}$ is the control vector. Standard errors cluster at the own-team level. The three binary outcomes (*Scores*, *Any yellow*, *Substituted*) are estimated via a linear probability model so that all coefficients read as percentage-point changes.

Cross-context regression (Table 3). For each domestic outfield player observed in both his domestic top-flight league and his own national team, we estimate:

$$Y_{p,m,t} = \beta \, \mathrm{NT}_m + X'_{p,m,t}\gamma + \alpha_p + \delta_t + \epsilon_{p,m,t} \tag{2}$$

In equation (2), NT_m is *National-team match*; player fixed effects α_p absorb time-invariant individual ability, and season fixed effects δ_t absorb season-specific shocks common across players; $X'_{p,m,t}$ is the control vector. Standard errors cluster at the player level. Because each player is observed in both contexts, the within-player design lets β read as the same player's performance shift between league and national team, controlling for his fixed ability.

Substitution-decision regression (Table 4). The manager's replacement choice is itself the behavior we want to read: when a foreign player is withdrawn, who comes on? To explore this, we fit, for each of three replacement-choice outcomes:

$$Y_e = \beta\, D_e + X_e'\gamma + \alpha_j + \delta_t + \epsilon_e \tag{3}$$

In equation (3), e indexes substitution events; D_e is the indicator *Withdrawn player is foreign*; α_j and δ_t are team and season fixed effects; and X_e' is the control vector. Standard errors cluster at the team level. Because withdrawal and replacement are simultaneous manager choices, β captures the manager's revealed behavioral preference.

Goal-creating chains (Figure 2). The third dimension we examine descriptively rather than by regression: for each goal-creating chain, we decompose the sequence by the nationality of the player occupying each position, from the start of the build-up to the scorer, and compare the resulting composition across the three leagues. Goal-creating chains are sequences of passes, and we follow Grohsjean, Piezunka, and Mickeler (2025) in reading within-team passing as a behavioral trace of teammate collaboration.[4]

[4]They operationalize collaboration as directed pass counts within teammate dyads; we borrow the measurement philosophy rather than that design, reading chain composition rather than dyadic intensity.

INITIAL DISCOVERY

Foreign players drove match victories in all three leagues, but the CSL effect was largest by a wide margin. Figure 1 reports country-specific slopes of match-win probability on two measures of foreign-player advantage. Each one-starter increase in the own team's foreign-starter advantage raised match-win probability by 5.52 pp in CSL ($p < .01$), versus 1.38 in J1 ($p < .05$) and 0.58 in K1 (ns). On the value-weighted measure CSL was again largest (3.08 pp per log-value unit, $p < .01$), with significant effects in J1 (0.90, $p < .01$) and K1 (2.04, $p < .05$) as well.

Foreign-player advantage predicted winning in all three leagues, and CSL emerged as the strongest case on both measures. This sharpens our question: when the foreign players who drive winning are unavailable, what happens to their domestic teammates?

Insert Figure 1 about here

To examine how domestic players responded, we turned to within-match foreign-injury shocks. Table 2 reports player-match regressions of seven domestic-player outcomes (dribble success, passing accuracy, scoring probability, yellow-card rate, substitution probability, on-ball touches, and overall match rating) on two injury shocks entered jointly: the net value lost to foreign-player absences and the net count of foreign players absent. Value and count, though correlated, are conceptually separable measures of what a team loses; each panel covers one league.

Yet where foreign players mattered most for winning, their domestic teammates' own in-match performance changed little in their absence. In CSL, the value-weighted shock showed no significant association with any of the seven outcomes, and the count-weighted shock produced only modest associations on passing accuracy (-0.90 pp) and scoring probability (+1.12 pp, both $p < .05$). In J1 the two shocks together moved four of the seven outcomes (scoring, substitution, touches, and rating), and in K1 the value shock moved passing accuracy and rating. For the CSL value-weighted shock, where no association is significant, the estimates are tightly bounded: for each unit of foreign value lost, equivalent to a single absent foreign player worth roughly €1.7 million, the 95% confidence interval rules out a passing-accuracy effect beyond roughly one percentage point. We report these intervals descriptively; the full set of confidence-interval and power bounds for all seven outcomes is provided in our response to reviewers.

All three leagues import foreign players under broadly similar quota regimes, yet the role those players occupy appears to differ in ways the player-match analysis only begins to reveal. Before turning to event-level evidence, we examined one further within-player contrast that holds individual capability fixed by construction: the same domestic player's performance when relocated to his senior national team, a setting that contains no foreign teammates.

Insert Table 2 about here

Table 3 reports within-player fixed-effect estimates for China, Japan, and Korea, in each case stacking dual-context players' club-league and national-team matches into one panel. The dependent variables are the seven outcomes from Table 2. Player fixed effects absorb time-invariant individual heterogeneity; season fixed effects absorb common time shocks. The *National-team match* coefficient identifies the within-player level change as the same player moves between the two settings, his club league and his national team, with his time-invariant ability differenced out.

Across the three leagues, the same domestic player's passing accuracy moved in opposite directions when he relocated from his club league to his national team: it fell in China (-4.95 pp, $p < .01$) but rose in Japan (+4.36 pp) and Korea (+2.81 pp, both $p < .01$). These between-country differences are themselves significant: a pooled interaction places China apart from both Japan and Korea ($p < .01$ for each contrast), with no significant difference between Japan and Korea (Figure 4(a)). Touches and overall match rating also fell in China; rating rose in Japan; dribble success rose in Korea. We read these within-player level changes as one signal among several, not as a verdict on national-team capability per se: many features differ between league and national-team contexts (training, tactical systems, opposition quality, teammate composition, role assignment), and player and season fixed effects absorb only those that are individual- or time-invariant. What is harder to attribute to such differences is the cross-country reversal in direction: moving the same player to a national-team setting with no foreign teammates was associated with his performance falling in China while rising in Japan and Korea, a pattern that mirrors at the within-player level the cross-country divergence in foreign–domestic dynamics we noted at the match level (Figure 1) and the player-match level (Table 2).

Insert Table 3 about here

SUBSEQUENT DISCOVERY

Where Tables 2 and 3 explored how domestic-player performance responded to foreign-teammate absence and across club versus national-team contexts, we next examined the moment when a foreign teammate exits the field mid-match. Who walks on in their place, and from what position? Table 4 reports linear probability models on 51,700 substitution events across CSL (2015–2025), J1 (2017–2025), and K1 (2020–2025). The key regressor is an indicator for whether the withdrawn player was foreign; outcomes capture whether the incoming replacement is a foreign player, a domestic forward, or in a position more defensive than the outgoing player's.

When a foreign player exited the field in a CSL match, the manager's response was distinctive: another foreign player came on, or the team retreated toward defense. CSL managers replaced foreign with foreign at 14.20 pp above the baseline ($p < .01$), versus 2.61 in J1 (ns) and a significant effect in K1 (6.73, $p < .01$). Yet the break runs between CSL and its neighbors: CSL differs from both J1 and K1 at $p < .01$, while K1, despite its own non-zero effect, does not differ significantly from J1 (Figure 4(b)). This elevated propensity ran against, not with, foreign-player availability: the unconditional foreign-substitute rate was *lower* in CSL (12%) than in J1 (18%) or K1 (20%). The foreign-for-foreign response thus reflects how CSL managers allocated a scarcer foreign pool rather than a deeper one to draw from. J1 and K1 managers promoted a domestic forward into the vacated role at roughly twice the CSL rate (12.08 and 10.56 pp, respectively, versus CSL's 6.44; all $p < .01$); and CSL managers were three times more likely than J1 or K1 managers to retreat toward a more defensive position when a foreign player exited.

The CSL foreign-replacement effect held under stringent tests of managerial discretion. Restricting the sample to injury-triggered substitutions left it at 12.71 pp ($p < .01$), nearly twice the J1 estimate; replacing team fixed effects with manager fixed effects (162 unique CSL managers, 100 in J1, 65 in K1) left it essentially unchanged at 14.22 pp ($p < .01$). What survives these tests is a revealed allocation preference, sharply divided across the three leagues at the moment a foreign player exits the field. In CSL, the manager either reaches for another foreign player or retreats the team toward defense; a domestic substitute, if brought on at all, is typically redirected to a more defensive position rather than promoted into the vacated role. In J1 and K1, by contrast, the same exit prompts the manager to promote a domestic forward into the offensive role. The CSL pattern reflects two interlocking layers of management decision-making. At the squad-construction layer, clubs concentrate foreign players in the team's most influential on-field positions. At the in-match substitution layer documented above, managers reproduce that allocation; the regularity is shared across the league's managerial population rather than driven by individual managerial philosophies. Within CSL, this foreign-for-foreign response is uniform across foreign-player origin, career stage, and team competitive pressure (Appendix Figure A.1), consistent with a league-wide schema rather than a tactic activated only under title or relegation pressure. Whether the manager-level regularity also shapes how players distribute the ball, and whether it leaves a geometric trace on the pitch, are the questions we examined next.

Insert Table 4 about here

FINAL DISCOVERY

Where Table 4 read the manager's hand at the moment of substitution, the goal-creating chains show where the play runs once it is in motion. Two descriptive exhibits trace that geometry: Figure 2 shows the composition of goal-creating chains by length and depth, and Figure 3 maps the spatial distribution of foreign-player on-ball activity across the pitch.

Foreign players dominate every position of CSL goal-creating chains, from deep in the build-up to the scorer. Chain lengths are similar across the three leagues, so cross-country differences lie in composition rather than length. The scorer is foreign in roughly 63% of CSL chains, against 21–29% in J1 and 36–47% in K1. Even at the deepest position in a five-touch chain (the start of the build-up), the player is foreign in 45% of CSL chains, against 15% in J1 and K1. These shares sit well above the foreign players' own share of the lineup, so the concentration reflects how chains are built rather than merely how many foreign players take the pitch.

Insert Figure 2 about here

Foreign and domestic players occupy more divergent regions of the pitch in CSL than in J1 or K1. Figure 3 plots the foreign-player share of minute-weighted on-ball events as a function of pitch position (F-share $= 0.5$ marks spatial overlap; deviations measure divergence). The total variation distance between the foreign and domestic activity distributions is 0.13 in CSL, 0.11 in K1, and 0.06 in J1: CSL and K1 both concentrate foreign activity in narrower, attacking regions while domestic activity spreads across the rest, whereas J1 players share the pitch most evenly. On this spatial measure, the highest- and lowest-spending leagues *deploy* their foreign players in similarly concentrated attacking regions, and here Korea resembles China more than Japan. But this spatial deployment, *where* foreign players are positioned, is the substrate, not the architecture. The architecture is the *behavior* built atop it: the managerial protection of Table 4 and the foreign-run goal-creating chains of Figure 2. On both, Korea does not follow China: it does not differ significantly from Japan in substitution or in the national-team contrast, and differs only mildly from Japan in its chains. Foreign players reach the scorer far more often in CSL than in K1; because the two leagues share the same concentrated deployment, that gap reflects how foreign players are used, not where they are positioned. Korea, in other words, shares China's concentrated spatial deployment but not its foreign–domestic talent gap, which (as we develop below) is as narrow in Korea as in Japan but several times wider in China; concentrated deployment alone does not produce the architecture.

Insert Figure 3 about here

DISCUSSION

Theory Contribution

A coherent architecture emerges across our analyses. CSL domestic players' own in-match performance changed little when their foreign teammates were absent (Table 2). The peripheral role they occupy appears to reflect how on-field responsibilities are allocated, rather than a ceiling on domestic capability. Our contribution lies not in documenting this stability but in uncovering the active arrangement built around it. Foreign players occupy central offensive roles while domestic players provide peripheral support, sustained by both managerial allocation and the foreign-centered routing of goal-creating play.[5] When foreign players exit the pitch, CSL managers substitute foreign-for-foreign rather than promote a domestic forward, and the goal-creating play runs through foreign players. We describe this pattern as a foreign-centered architecture: a concentra-

[5] Defensive contribution can be disproportionately important to team outcomes (Groysberg, Lin, Naik, & Schmidt, 2023); the architecture's consequentiality, in this reading, stems from its offensive over-concentration rather than from any tactical efficiency it might secure.

tion around foreign players rather than diversification through them. The pattern runs counter to European football evidence, where Glennon et al. (2025) document that skilled immigrants broaden club strategic variety. Within East Asia, the architecture is distinctive to CSL: J1 and K1 do not reproduce it (Figure 4), even though K1, as shown above, deploys foreign players in similarly concentrated attacking regions. Figure 4 collects these cross-country contrasts: each panel plots the three between-league differences for one outcome, with 95% confidence intervals.

Insert Figure 4 about here

Why does the architecture take hold in China but not in its neighbors? Our reading of the evidence is that it requires two conditions to coincide, neither sufficient alone. The first is the talent gap. Only when foreign players arrive as stars, far above their domestic teammates rather than near-equals, does it become worthwhile to build the team around them and to protect that arrangement; this is why the pattern is a break rather than a smooth function of spending. The second condition is institutional: CSL clubs face acute win-now pressure that rewards leaning on the foreign core now over the slower work of exploration. Korea shows that both conditions are needed. Korean clubs concentrate their foreign players in similarly narrow attacking regions, but because their talent gap is as narrow as Japan's, the architecture never forms. Where both hold, the architecture feeds on itself: each win secured through the foreign core raises the reward for doing it again. With only three leagues we cannot statistically separate these conditions, so we read the cross-country comparison as consistency evidence; a complementary within-CSL test points the same way, with the architecture strengthening as the talent gap widens across team-seasons

(Table A.1). Because the gap is itself a team-season choice rather than an assigned treatment, we read this within-CSL gradient as an association, not a causal estimate.

We develop these arguments through three established literatures. The talent-gap argument draws on studies of stars and high-performer peer effects; the institutional-pressure argument draws on the exploration–exploitation literature. Working-partnership durability (from the social capital and human capital resource literatures) is a within-CSL mechanism that does not, as we explain, travel across leagues, and so is not one of the two cross-country conditions. We take the two cross-country conditions first, beginning with the talent gap, and then turn to the within-CSL mechanism.

The Talent Gap and the Star Role

The first cross-country condition operates at the team level: the foreign–domestic talent gap. Figure 5(a) shows the gap directly: the foreign-to-domestic peak market value ratio of appearance-weighted medians is 11.2 in CSL (an order of magnitude) against just 1.7 in both J1 and K1. Where this gap is wide, foreign and domestic players are far from interchangeable, and foreign players are received more as stars than as peers. Drawing on O'Boyle and Götz (2026), we treat such foreign players as the relative top of a shallow talent pool, exerting a "gravitational" pull on attention and resources.[6] Asgari et al. (2021) caution that stars do not always deliver expected returns and may instead produce "nonoccurrence," crowding out contributions nonstars might otherwise have made. When stars concentrate in central offensive roles while nonstars provide peripheral support, the resulting architecture evokes Casciaro and Piskorski (2005)'s twin constructs of power

[6] We use "star" relativistically rather than in the absolute-superstar formulations of Rosen (1981), who describes small talent differences magnified by imperfect substitution and scale economies of joint consumption, and Adler (1985), who focuses on stardom emerging from consumer coordination even without talent differences. Franck and Nüesch (2012) show empirically that both mechanisms operate in elite soccer. See Call et al. (2015) for an integrative review reconciling the economics, sociology, and management star literatures.

imbalance and mutual dependence. Building on those constructs, Kehoe and Tzabbar (2015: 714) describe such a setting as one in which stars may adopt a "directive and protective stance" that limits tacit-knowledge sharing. At the individual level, Flynn and Amanatullah (2012) offer a complementary mechanism: a high-status coactor elevates a focal actor's performance under independent coaction but suppresses it under direct competition over a single allocable reward, as the focal actor's expectancy of capturing the reward diminishes. In the CSL setting, where central offensive roles, ball-touches in creation lanes, and managerial attention behave as scarce, foreign-anchored rewards rather than as independently allocated opportunities, this mechanism, applied to the domestic player, predicts the direction we observe. Upstream of these allocation dynamics, Terry, McGee, Kass, and Collings (2023: 320) argue that acquirers tend to discount the team-level and institutional conditions that originally produced a star's visible performance, yielding a winner's-curse dynamic in compensation that we argue subsequently anchors role allocation.

Insert Figure 5 about here

Tzabbar and Kehoe (2014) find that star exit creates opportunity for peer exploration primarily when the departing star's innovative involvement in the firm was low; when involvement was high, the post-exit exploration upside is dampened. CSL foreign players fit the high-involvement profile, and the post-departure pattern we observe is consistent with such dampening: replacement remains foreign-for-foreign, and the central offensive roles continue to flow through foreign players. Cross-country, the star dynamic identified by Rosen (1981) is more strongly engaged where the gap is wide; where it is narrow, domestic players are closer substitutes and foreign players are received more as peers than as stars. This distinction predicts a weaker pattern in the narrow-gap leagues, where comparable status invites integration rather than deference and the architecture does not take hold. This fits the Korean case: the foreign-for-foreign substitution signature is milder and does not differ significantly from Japan's (Figure 4), and foreign players run the goal-creating chains far less than in CSL (Figure 2). Within CSL, a wider team-season gap further amplifies the substitution signature: a one-standard-deviation increase in the team's log foreign-to-domestic market-value ratio strengthens the foreign-for-foreign effect by 4.91 pp (Table A.1, column 1, $p < .01$), while leaving the defensive-shift placebo unmoved (column 2), indicating that the substitution and defensive channels operate through independent mechanisms rather than a common response to roster imbalance. This reading aligns with Hendricks et al.'s (2023: 1997) synthesis of high-performer peer effects, in which those effects are conditional on "the characteristics of high performers, peers, and the context" in which they interact.

Institutional Pressures and Exploration Tolerance

The second cross-country condition operates at the institutional level: the shared pressure on clubs within a league to favor exploitation over exploration. Because this pressure is by construction common across clubs within a league, it cannot be tested through the team-season moderation we applied to the talent gap; we read its operation through two complementary signatures. March (1991: 71, 73) identifies the source of this pressure: returns from exploration are "less certain, more remote in time, and organizationally more distant from the locus of action" than returns from exploitation, so organizations entrench established routines faster than they explore new ones and may become "self-destructive in the long run." Competence at the foreign core compounds: each gain raises the reward for relying on it again, a self-reinforcing logic that can come to dominate when the institutional levers favor short-cycle returns.

Levinthal and March (1993: 95) sharpen this dynamic into three forms of learning myopia: "the tendency to overlook distant times, distant places, and failures." Temporal myopia operates within clubs: under short-horizon pressures from sponsors, standings, and regulatory uncertainty, longer-horizon returns are discounted heavily against immediate wins driven by foreign players. Spatial myopia operates across the league: an individual organization may rationally choose to exploit others' successful explorations rather than to explore independently, so that when all clubs follow this strategy, the system as a whole leans toward exploitation. Sydow, Schreyögg, and Koch (2009) describe how such patterns may crystallize into organizational path dependence, with self-reinforcing mechanisms producing lock-in that resists redirection even when context shifts. That the architecture appears in CSL alone is consistent with variation in what Levinthal and March (1993: 107–109) identify as the levers that keep exploration alive against the pull toward exploitation: incentives, organizational structure, beliefs, and internal selection. CSL clubs face acute championship-race and regulatory pressures that compress payoff horizons and reward exploitation over the longer cycles exploration would require; J1 and K1 clubs, operating within longer-tenured ownership and more established youth academies, may face weaker pressures and retain greater tolerance for experimentation.

Two empirical signatures support this condition. First, the architecture documented in Table 4 survives the absorption of 162 unique CSL manager fixed effects (column 5): it does not depend on which manager is in charge, but operates as a shared schema across the managerial population. Such field-level convergence reflects the institutional isomorphism DiMaggio and Powell (1983) attribute to shared regulators, a common labor market, and imitation among clubs. It points to a common response to shared incentives, not a distinctively Chinese coaching culture or any individual coaching philosophy: the managers are responding to the conditions they face. Second, managerial turnover is highest in the CSL: clubs there average just 1.12 club-seasons per manager, against 1.37 in K1 and 1.74 in J1, alongside a CSL regulatory environment that cycled through transfer taxes, salary caps, and evolving foreign-quota and U23 mandates over the sample window. Under this reading, the role imported talent comes to play varies with the institutional conditions that determine how much exploration tolerance each league sustains.

Working-Partnership Durability and Capability Emergence

Beyond these two cross-country conditions, a within-CSL mechanism operates at the dyadic level: the durability of foreign–domestic working partnerships. Ployhart et al. (2014: 393) distinguish individual knowledge from unit-level human capital resources, which emerge through interaction and interdependence over time and which, they argue, have "no factor market." Reagans, Argote, and Brooks (2005) identify experience working together as a distinct, dyadic learning asset that accumulates as the same individuals coordinate over time. They also describe a "negative transfer" hazard, in which relationship-specific knowledge built with prior teammates can be misapplied to a new set of teammates.

Nyberg and Ployhart (2013) theorize that in intensive-workflow settings, the unit-performance consequences of collective turnover depend on both the quality of the capital lost and whether replacement inflows match it in quality and quantity. Short foreign-player contracts, frequent transfer activity, and cross-cultural barriers to the kind of shared codes and language Nahapiet and Ghoshal (1998) identify as constituting the cognitive dimension of social capital are conditions that act on the same accumulation channel. Loignon et al. (2025) document the empirical signature in European top-league soccer: low-density, high-centralization hub-and-spokes configurations are associated with the lowest performance among talent-heavy teams, a signature the CSL goal-creating chains bear. Such conditional dynamics resonate with Campbell et al. (2017)'s integration of social-comparison and conservation-of-resources theories, in which peers' responses to high performers reflect a deliberate resource calculus over and above emotional reactance. When foreign players occupy preferred positions and absorb the finite resource pool of ball-touches and managerial attention, the foreign concentration of goal-creating chains we document in Figure 2 is consistent with resource-protective behavior at the dyadic level. Cross-league, however, foreign-player tenure does not track the architecture. Korea's foreign players turn over fastest of the three (median club tenure of 179 days, against 217 in CSL and 251 in J1; Figure 5), yet Korea does not reproduce the pattern: in managerial substitution the CSL–K1 contrast is significant ($p < .01$), while K1 and J1 do not differ significantly (Figure 4). We therefore read partnership durability not as a cross-country driver but as a within-CSL condition, one repeatedly reset by the boom-and-bust churn of the league's import market. From this perspective, the role imported talent comes to play varies with whether the team-level interactions through which shared resources accumulate are sustained or repeatedly reset.

Limitations and Future Research

Despite these contributions, our study features several limitations. First, commercial event-data coverage of Asian top-flight football remains substantially less developed than coverage of European competitions in both the dimensions captured and the historical depth per dimension. For comparison, Glennon et al. (2025) draw on match outcome and lineup data for the European Big Five leagues across a thirty-year window (1990–2020), with finer-grained pass-level (play-by-play) coverage only from the 2009–2010 season; analogous depth for CSL, J1, and K1 is only now beginning to accumulate. Our study sits at the current frontier of what major commercial vendors release for these three Asian leagues, and our goal-creating-chain analysis, in particular, became feasible only with the 2024–2025 expansion of Sofascore's chain coverage. As commercial providers continue to expand the breadth and granularity of their Asian-football coverage, and as historical depth accumulates with each successive season, future research will be able to extend the cascade we document across longer time windows and additional event dimensions, allowing sharper tests of whether the architecture is stable across policy cycles or shifts with each successive regulatory wave.

Second, our design carries two interpretive limits. CSL was subject to multiple overlapping regulatory regimes during our sample window: transfer taxes, salary caps, foreign-player quotas and U23 mandates, and integrity reviews. To our knowledge, because these policies overlap in time and apply uniformly across all clubs, no current causal-inference technique can decompose this bundle into the marginal contribution of each individual instrument. Relatedly, because our outcomes draw entirely from match event data, the within-club training environment lies outside our observational window; our findings accordingly describe foreign–domestic match-day patterns rather than what unfolds in day-to-day preparation. If the match-day configuration we document reflects underlying training-ground practices, then the role allocations we observe on the pitch may also be present in day-to-day team operations; ethnographic or field-investigation studies with access to club-internal training data could test this conjecture directly. Comparative analysis of analogous regulatory bundles in other leagues could in turn triangulate the response pattern across different policy mixes.

Finally, although sport offers unusually precise and comprehensive data for management research (Fonti, Ross, & Aversa, 2023), the scope of what we document warrants care. We do not claim that foreign-talent importation is broadly harmful; often it complements local participants and makes room for them rather than crowding them out. What we document is the configurational condition under which displacement, rather than complementation, prevails when imported labor is concentrated in the same organizational roles that local participants would otherwise occupy. Analogous patterns of displacement and crowd-out have been documented in high-skilled immigration settings (Borjas & Doran, 2012; Doran, Gelber, & Isen, 2022). Whether this configurational cascade governs other talent-import contexts (skilled-immigrant R&D, returnee-scientist programs, expatriate manager rotation in multinational corporations) is a question for future research beyond sport.

Practical Implications

Our findings carry one principle-level implication for managers of talent-import settings: role configuration matters more than headcount. The broader literature on talent management has repeatedly shown that the within-team configuration of high-status individuals conditions whether and how their numbers translate into group outcomes (Groysberg, Polzer, & Elfenbein, 2011; Loignon et al., 2025). The cascade we document, from squad construction through substitution to the goal-creating chains, suggests that the role imported talent comes to play hinges on how foreign and domestic players share organizational responsibilities, not on the volume of foreign players. Where foreign players concentrate in the same organizational roles that local participants would otherwise occupy, the architecture becomes self-reinforcing. Adjusting the number of foreign players without restructuring this role-sharing is therefore unlikely to alter the architecture.

Consistent with this principle, Groysberg et al. (2023) observe that once a league has sunk large fees into marquee imports, those players must keep playing and the room for developing players narrows, a dynamic that parallels the architectural lock-in we document in the CSL. Cross-group collaboration rates, position-allocation concentration, and substitution patterns when foreign players exit can serve as early indicators of architectural lock-in.

CONCLUSION

Our central discovery is a coherent foreign-centered architecture: in the Chinese Super League, foreign players hold the central offensive roles while domestic players are confined to peripheral support, produced by managerial allocation and reflected in the foreign-centered routing of goal-creating play. It is distinctive to CSL; J1 and K1 do not reproduce it, even though Korea deploys foreign players in similarly concentrated attacking regions. We read it as sustained by two cross-country conditions (a wide foreign–domestic talent gap that casts foreign players as stars, and institutional win-now pressure that narrows tolerance for experimentation), together with a within-CSL mechanism, the durability of foreign–domestic partnerships. The deeper lesson is that whether imported talent complements or displaces local participants turns not on quantity but on placement: when imports occupy the central roles locals would otherwise fill, they crowd them out rather than make room for them. Whether the same dynamic operates beyond sport, from skilled-immigrant R&D to expatriate management, is a question we leave open. In most organizations, who occupies the central roles and how well they perform cannot be traced as finely as on a football pitch; that rare observability is what let us see this architecture, and what makes sport a revealing place to study it.

References

Adler, M. 1985. Stardom and talent. ***American Economic Review***, 75: 208–212.

Asgari, E., Hunt, R. A., Lerner, D. A., Townsend, D. M., Hayward, M. L. A., & Kiefer, K. 2021. Red giants or black holes? The antecedent conditions and multilevel impacts of star performers. ***Academy of Management Annals***, 15: 223–265.

Borjas, G. J., & Doran, K. B. 2012. The collapse of the Soviet Union and the productivity of American mathematicians. ***Quarterly Journal of Economics***, 127: 1143–1203.

Call, M. L., Nyberg, A. J., & Thatcher, S. M. B. 2015. Stargazing: An integrative conceptual review, theoretical reconciliation, and extension for star employee research. ***Journal of Applied Psychology***, 100: 623–640.

Campbell, E. M., Liao, H., Chuang, A., Zhou, J., & Dong, Y. 2017. Hot shots and cool reception? An expanded view of social consequences for high performers. ***Journal of Applied Psychology***, 102: 845–866.

Casciaro, T., & Piskorski, M. J. 2005. Power imbalance, mutual dependence, and constraint absorption: A closer look at resource dependence theory. ***Administrative Science Quarterly***, 50: 167–199.

DiMaggio, P. J., & Powell, W. W. 1983. The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. ***American Sociological Review***, 48: 147–160.

Doran, K., Gelber, A., & Isen, A. 2022. The effects of high-skilled immigration policy on firms: Evidence from visa lotteries. ***Journal of Political Economy***, 130: 2501–2533.

Flynn, F. J., & Amanatullah, E. T. 2012. Psyched up or psyched out? The influence of coactor status on individual performance. ***Organization Science***, 23: 402–415.

Fonti, F., Ross, J.-M., & Aversa, P. 2023. Using sports data to advance management research: A review and a guide for future studies. ***Journal of Management***, 49: 325–362.

Franck, E., & Nüesch, S. 2012. Talent and/or popularity: What does it take to be a superstar? ***Economic Inquiry***, 50: 202–216.

Glennon, B., Morales, F., Carnahan, S., & Hernández, E. 2025. Does employing skilled immigrants enhance competitive performance? Evidence from European football clubs. ***Management Science***, 71: 5746–5766.

Grohsjean, T., Piezunka, H., & Mickeler, M. 2025. When colleagues compete outside the firm. ***Strategic Management Journal***, 46: 640–666.

Groysberg, B., Lin, E., Naik, A., & Schmidt, S. L. 2023. Identify critical roles to improve performance. ***MIT Sloan Management Review***, 64(4): 1–5.

Groysberg, B., Polzer, J. T., & Elfenbein, H. A. 2011. Too many cooks spoil the broth: How high-status individuals decrease group effectiveness. ***Organization Science***, 22: 722–737.

Hendricks, J. L., Call, M. L., & Campbell, E. M. 2023. High performer peer effects: A review, synthesis, and agenda for future research. ***Journal of Management***, 49: 1997–2029.

Hvattum, L. M., & Arntzen, H. 2010. Using ELO ratings for match result prediction in association football. ***International Journal of Forecasting***, 26: 460–470.

Kehoe, R. R., & Tzabbar, D. 2015. Lighting the way or stealing the shine? An examination of the duality in star scientists' effects on firm innovative performance. ***Strategic Management Journal***, 36: 709–727.

Levinthal, D. A., & March, J. G. 1993. The myopia of learning. ***Strategic Management Journal***, 14: 95–112.

Li, Y., Li, N., Li, C., & Li, J. 2020. The boon and bane of creative "stars": A social network exploration of how and when team creativity is (and is not) driven by a star teammate. ***Academy of Management Journal***, 63: 613–635.

Loignon, A. C., Fonti, F., Bagherzadeh, M., Gurca, A., & Shrestha, S. 2025. When more is less: The role of social capital in managing talent in teams. ***Academy of Management Discoveries***, 11: 255–284.

March, J. G. 1991. Exploration and exploitation in organizational learning. ***Organization Science***, 2: 71–87.

Mas, A., & Moretti, E. 2009. Peers at work. ***American Economic Review***, 99: 112–145.

Nahapiet, J., & Ghoshal, S. 1998. Social capital, intellectual capital, and the organizational advantage. ***Academy of Management Review***, 23: 242–266.

Nyberg, A. J., & Ployhart, R. E. 2013. Context-emergent turnover (CET) theory: A theory of collective turnover. ***Academy of Management Review***, 38: 109–131.

O'Boyle, E. H., & Götz, M. 2026. Rethinking stardom: A relativistic approach to studying the absolute best performers. ***Annual Review of Organizational Psychology and Organizational Behavior***, 13: 107–136.

Panja, T., Chang, R., & Browning, J. 2016. July 10: Chinese billions flood soccer, snaring Hulk in record deal. ***Bloomberg***. Retrieved from https://www.bloomberg.com/news/articles/2016-07-10/chinese-billions-flood-into-soccer-snaring-hulk-in-record-deal.

Ployhart, R. E., Nyberg, A. J., Reilly, G., & Maltarich, M. A. 2014. Human capital is dead; long live human capital resources! ***Journal of Management***, 40: 371–398.

Reagans, R., Argote, L., & Brooks, D. 2005. Individual experience and experience working together: Predicting learning rates from knowing who knows what and knowing how to work together. ***Management Science***, 51: 869–881.

Rosen, S. 1981. The economics of superstars. ***American Economic Review***, 71: 845–858.

RTÉ Sport 2016. June 30: Shanghai stump up €55.8 million for Hulk. Retrieved from https://www.rte.ie/sport/soccer/2016/0630/799250-shanghai-sipg-for-a-record-55-8-million/.

Sky Sports 2016. December 29: Carlos Tevez moves to China and becomes world's highest paid player. Retrieved from https://www.skysports.com/football/news/11095/10710809/carlos-tevez-moves-to-china-and-becomes-worlds-highest-paid-player.

Sydow, J., Schreyögg, G., & Koch, J. 2009. Organizational path dependence: Opening the black box. ***Academy of Management Review***, 34: 689–709.

Terry, R. P., McGee, J. E., Kass, M. J., & Collings, D. G. 2023. Assessing star value: The influence of prior performance and visibility on compensation strategy. ***Human Resource Management Journal***, 33: 307–327.

Twomey, L. 2016. December 23: Chelsea announce £60 million Oscar transfer to Shanghai SIPG. ***ESPN***. Retrieved from https://www.espn.com/soccer/story/_/id/37507224.

Tzabbar, D., & Kehoe, R. R. 2014. Can opportunity emerge from disarray? An examination of exploration and exploitation following star scientist turnover. ***Journal of Management***, 40: 449–482.

TABLES

TABLE 1
Summary Statistics

	China (CSL)		Japan (J1)		Korea (K1)	
	Domestic	Foreign	Domestic	Foreign	Domestic	Foreign
Player and match characteristics						
Age (years)	26.34	28.50	26.19	27.82	26.33	27.94
Market value (Peak, million EUR)	0.45	6.61	1.48	3.09	0.89	1.71
Starting lineup	46.97	83.69	60.96	64.47	59.62	61.70
Minutes played	42.55	74.38	54.95	58.02	52.84	58.22
Performance						
Dribble success	53.14	56.64	52.42	53.31	45.87	47.62
Pass success	77.68	77.54	77.69	74.79	78.62	74.62
Scores	3.63	19.37	5.93	12.07	5.28	13.94
Any yellow	9.32	12.20	6.25	10.12	9.56	10.76
Substituted	19.09	22.38	23.86	27.93	25.05	26.85
Touches	25.57	46.01	36.81	36.77	35.26	35.71
Rating	6.75	7.12	6.85	6.93	6.84	6.98
N (player-matches)	68,297	17,387	80,254	16,595	36,470	7,631
Unique players	1,094	493	1,217	357	781	230

Notes: Per-match means for outfield player-matches in China (CSL), Japan (J1), and Korea (K1), by foreign/domestic player status (goalkeepers excluded). *Starting lineup*, *Dribble success*, and *Pass success* are percentages (0–100). *Scores*, *Any yellow*, and *Substituted* are binary indicators reported in percentage points (0–100). *Touches* is a per-match count. *Rating* is the Sofascore composite performance rating, generated for players with at least 10 minutes of play.

TABLE 2

Domestic-Player Performance and Foreign-Teammate Injury Shocks

	(1) Dribble success	(2) Pass success	(3) Scores	(4) Any yellow	(5) Substituted	(6) Touches	(7) Rating
Panel A: China (CSL)							
Net Foreign Injury Loss: Value	-0.35	0.51†	-0.23	-0.59	-0.61	0.26	0.01
	(1.34)	(0.27)	(0.37)	(0.66)	(0.37)	(0.41)	(0.01)
Net Foreign Injury Loss: Count	-0.81	-0.90*	1.12*	0.53	0.02	0.29	-0.00
	(1.74)	(0.38)	(0.51)	(0.93)	(0.60)	(0.62)	(0.01)
Outcome mean	53.15	77.69	5.30	13.48	27.81	37.26	6.75
Observations	18,277	45,690	46,802	46,802	46,802	46,802	45,033
Panel B: Japan (J1)							
Net Foreign Injury Loss: Value	0.17	-0.63	-1.27*	-0.25	0.98*	1.43*	-0.01
	(1.19)	(0.39)	(0.57)	(0.48)	(0.48)	(0.69)	(0.01)
Net Foreign Injury Loss: Count	-0.25	0.39	1.02**	0.06	-0.60†	-0.52	0.02*
	(0.92)	(0.26)	(0.37)	(0.45)	(0.33)	(0.55)	(0.01)
Outcome mean	52.41	77.70	6.97	7.33	27.99	43.27	6.85
Observations	30,951	67,294	68,242	68,242	68,242	68,242	66,537
Panel C: Korea (K1)							
Net Foreign Injury Loss: Value	-1.34	-1.15**	-1.52†	-0.46	-0.04	-0.23	-0.03*
	(1.72)	(0.33)	(0.76)	(0.79)	(0.68)	(0.93)	(0.01)
Net Foreign Injury Loss: Count	1.74	0.32	0.91†	0.20	-0.29	-0.52	0.02†
	(1.24)	(0.26)	(0.47)	(0.52)	(0.56)	(0.66)	(0.01)
Outcome mean	45.88	78.64	6.27	11.30	29.58	41.87	6.85
Observations	13,735	30,276	30,695	30,695	30,695	30,695	29,072

Notes: Sample is domestic outfield players in China (CSL), Japan (J1), and Korea (K1) with positive minutes. *Net Foreign Injury Loss: Value* sums $\log(1 + \mathrm{MV}_i/1\mathrm{M})$ across the own team's injured foreign players i at the match date and subtracts the analogous opponent sum, where MV_i is Transfermarkt market value in euros; *Count* uses headcount in place of market value (positive values indicate a larger own-team shortfall). Outcome definitions match Table 1. All regressions include player and own-team $\times$ season fixed effects and controls for home indicator, opponent pre-match Elo, age, and age squared. Standard errors clustered by team are in parentheses.

$^{\dagger}\, p < .10$
$^{*}\, p < .05$
$^{**}\, p < .01$

TABLE 3

Performance in National Team versus Club

	(1) Dribble success	(2) Pass success	(3) Scores	(4) Any yellow	(5) Substituted	(6) Touches	(7) Rating
Panel A: China (CSL)							
National-team match	-0.11	-4.95**	48.32**	-3.69**	-2.99*	-4.36**	-0.18**
	(2.66)	(0.60)	(2.92)	(0.97)	(1.46)	(1.33)	(0.02)
Outcome mean	53.94	79.17	10.81	11.01	24.20	35.50	6.86
Observations	8,592	18,379	23,062	23,062	23,062	22,158	18,231
Panel B: Japan (J1)							
National-team match	3.80	4.36**	53.42**	-4.64**	-2.69	-3.41	0.10**
	(2.56)	(0.70)	(2.28)	(0.63)	(1.78)	(2.17)	(0.03)
Outcome mean	53.56	78.77	14.41	6.58	22.76	46.04	6.95
Observations	9,330	18,272	20,912	20,912	20,912	19,773	18,165
Panel C: Korea (K1)							
National-team match	8.20*	2.81**	47.82**	-6.24**	-3.53	-3.50	-0.05
	(3.28)	(0.80)	(2.61)	(1.33)	(2.52)	(2.54)	(0.03)
Outcome mean	47.91	80.69	13.22	10.37	23.32	46.13	6.95
Observations	5,214	10,454	12,086	12,086	12,086	11,409	10,271

Notes: Sample is dual-context domestic outfield players in China (CSL), Japan (J1), and Korea (K1) observed in both their domestic top-flight league and their own national team. *National-team match* is a binary indicator equal to one for national-team appearances and zero for league appearances. Outcome definitions match Table 1. All regressions include player and season fixed effects and controls for home indicator, opponent pre-match Elo, age, and age squared. Opponent Elo equals the club Elo for league matches and the international Elo for national-team matches. Standard errors clustered by player are in parentheses.

$^{*}\, p < .05$

$^{**}\, p < .01$

TABLE 4

Foreign-Centered Architecture in Manager Substitution Decisions

	(1) Foreign replacement	(2) Domestic forward replacement	(3) Defensive position shift	(4) Foreign replacement	(5) Foreign replacement
Panel A: China (CSL)					
Withdrawn player is foreign	14.20**	6.44**	14.10**	12.71**	14.22**
	(1.32)	(0.88)	(1.50)	(2.27)	(1.52)
Outcome mean	11.98	13.70	22.88	8.82	11.98
Observations	16,755	16,755	16,755	2,108	16,755
Panel B: Japan (J1)					
Withdrawn player is foreign	2.61	12.08**	4.44*	6.65*	2.72
	(2.08)	(1.58)	(1.98)	(2.79)	(1.81)
Outcome mean	17.97	21.89	19.69	14.39	17.97
Observations	23,814	23,814	23,814	1,640	23,814
Panel C: Korea (K1)					
Withdrawn player is foreign	6.73**	10.56**	4.73†	9.67†	6.74**
	(1.97)	(1.63)	(2.59)	(5.12)	(1.81)
Outcome mean	19.51	24.11	21.89	14.41	19.51
Observations	11,131	11,131	11,131	909	11,131
Team fixed effects	Yes	Yes	Yes	Yes	No
Manager fixed effects	No	No	No	No	Yes
Sample	Full	Full	Full	Injury-only	Full

Notes: Sample is one observation per in-game substitution event in China (CSL, 2015–2025), Japan (J1, 2017–2025), and Korea (K1, 2020–2025); pooled N=51,700 across 6,645 matches. *Withdrawn player is foreign* equals 1 if the outgoing player is foreign. Outcomes are in percentage points: *Foreign replacement* (columns 1, 4, 5) equals 100 if the incoming replacement is foreign; *Domestic forward replacement* (column 2) equals 100 if the incoming replacement is a domestic forward; *Defensive position shift* (column 3) equals 100 if the incoming position is more defensive than the outgoing (goalkeeper=0, defender=1, midfielder=2, forward=3). Column (4) restricts to mid-match injury-triggered substitutions; column (5) absorbs 162 (CSL), 100 (J1), and 65 (K1) unique manager fixed effects in place of team fixed effects. All specifications include season fixed effects and controls for substitution minute, injury indicator, running goal differential, and own-team foreign-starter count. Standard errors are clustered by team in columns (1)–(4) and by manager in column (5).

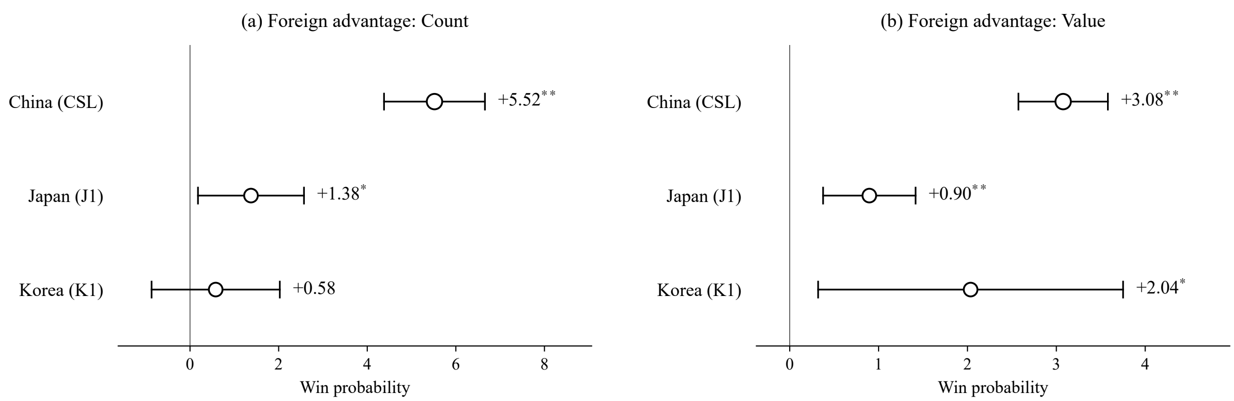
† $p < .10$

* $p < .05$

** $p < .01$

FIGURES

FIGURE 1
Match-Level Returns to Foreign-Player Advantage

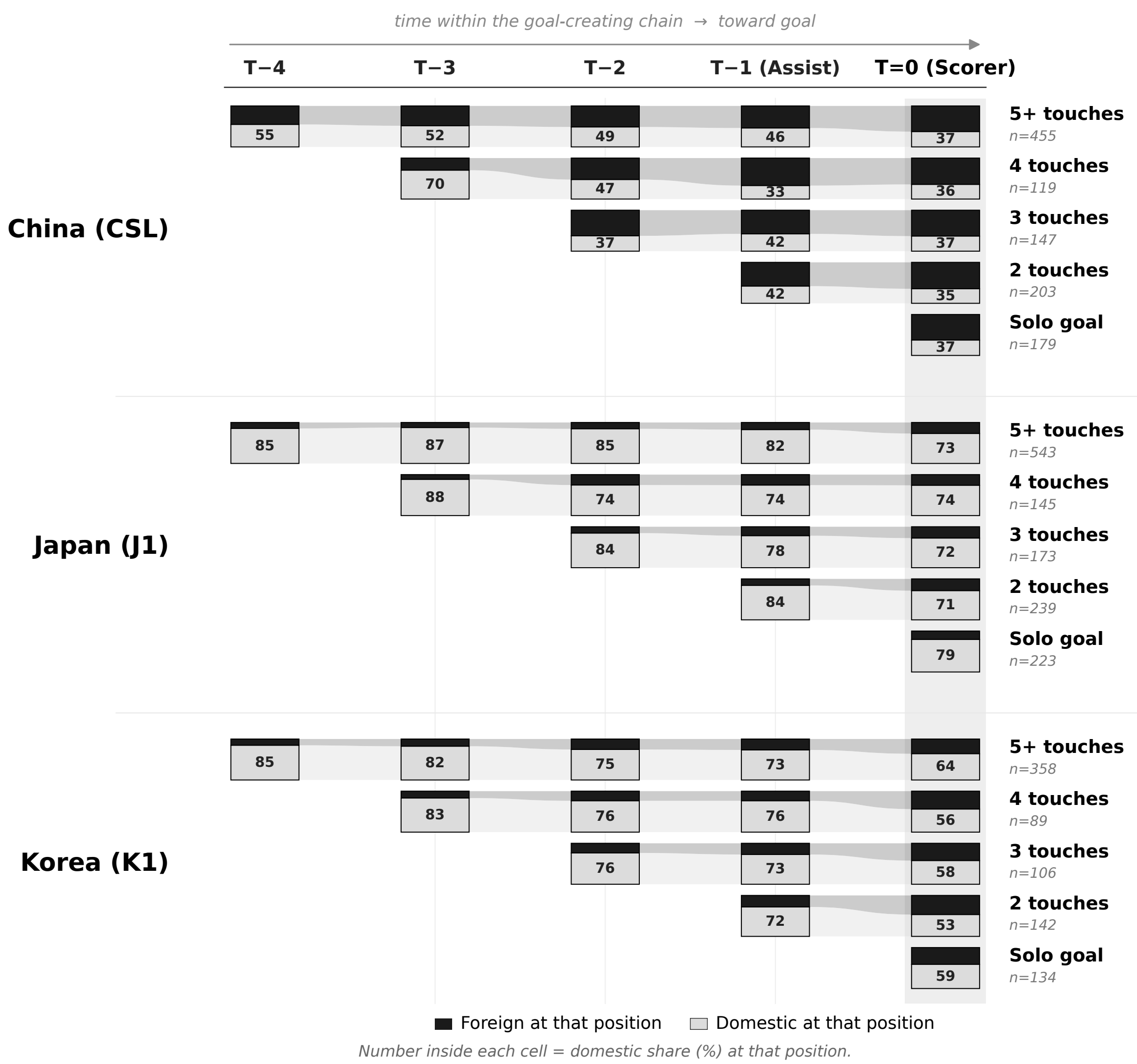


Notes: Sample is one observation per team per match in China (CSL, 2015–2025, $N = 4{,}680$), Japan (J1, 2017–2025, $N = 5{,}962$), and Korea (K1, 2020–2025, $N = 2{,}648$). *Foreign advantage: Count* is the own minus opponent count of foreign starters; *Foreign advantage: Value* is the analogous difference in the sum of log Transfermarkt market values. Each point estimate is a country-specific OLS slope of match-win probability (in percentage points) on the relevant advantage measure, with season fixed effects; standard errors clustered by team are in parentheses, and horizontal bars are 95% confidence intervals.

$^{*}\, p < .05$
$^{**}\, p < .01$

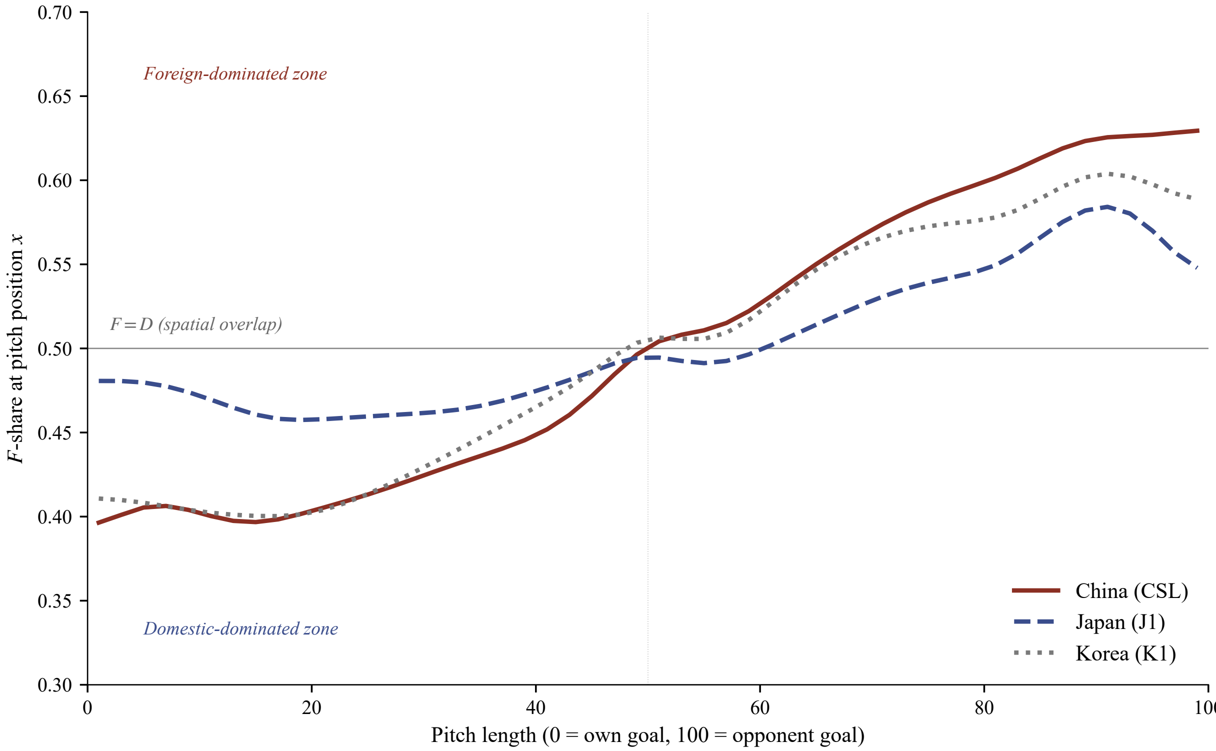
FIGURE 2

Foreign and Domestic Players in Goal-Creating Chains



Notes: Sample is 2024–2025 goal-creating chains in China (CSL, $N = 1{,}103$), Japan (J1, $N = 1{,}323$), and Korea (K1, $N = 829$). Chains are partitioned by number of touches into five lanes (1, 2, 3, 4, 5+); the 1-touch lane (“Solo goal”) captures unassisted shots and rebounds. Cells are right-aligned at the scorer ($T = 0$); each is a stacked bar split between foreign (black) and domestic (light gray) players, with the number inside reporting the *domestic* share (%).

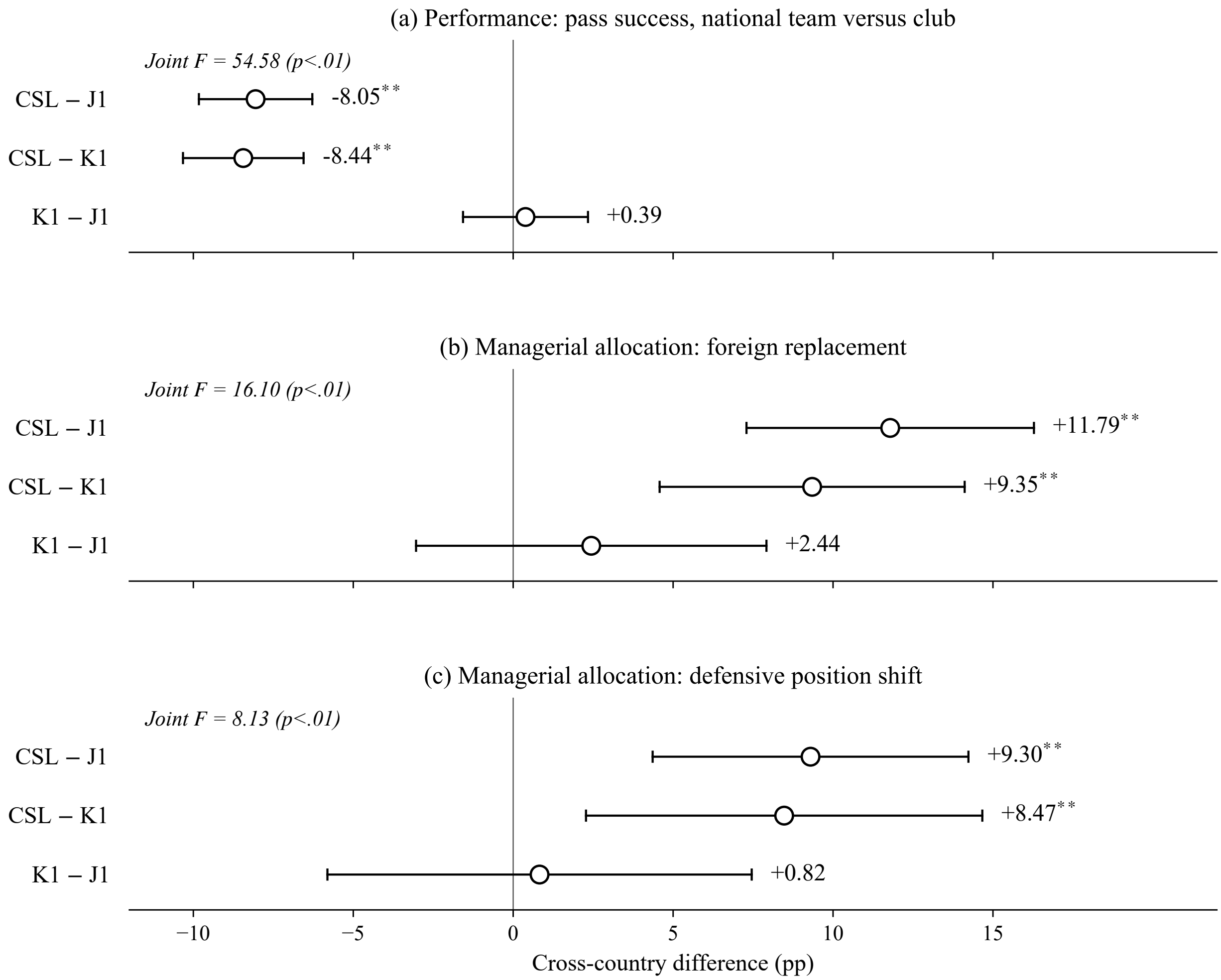
FIGURE 3
Foreign-Player Activity Share by Pitch Position



Notes: Sample is 7,583,273 outfield on-ball action events (touches, passes, tackles, etc.; goalkeepers excluded) from China (CSL, 2016–2025), Japan (J1, 2020–2025), and Korea (K1, 2020–2025), drawn from Sofascore's per-player heatmap. Each curve plots foreign players' share of minute-weighted events at pitch position x, F-share$(x) = p_F(x)/(p_F(x) + p_D(x))$, after 50-bin Gaussian smoothing ($\sigma = 2$ bins $\approx$ 4% of pitch); F-share $= 0.5$ marks foreign–domestic spatial overlap. Sample window reflects Sofascore per-player heatmap coverage and is narrower than the main analysis sample.

FIGURE 4

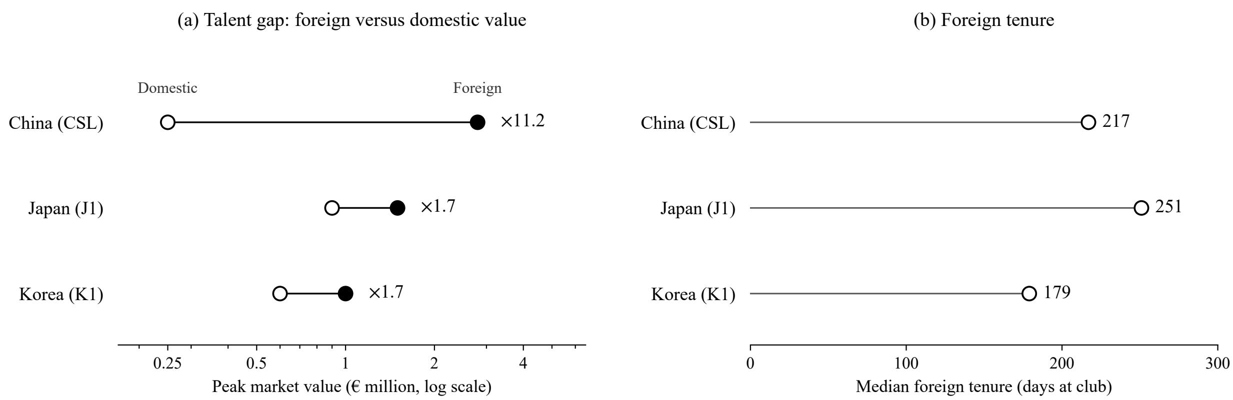
Cross-Country Contrasts in the Foreign-Centered Architecture



Notes: Each panel plots the three between-league contrasts (CSL−J1, CSL−K1, K1−J1) for one outcome, with 95% confidence intervals, from a pooled regression interacting the treatment with China and Korea indicators (Japan the omitted baseline) in that outcome's native specification; the joint F tests equality of the three leagues' effects. Panel (a): pass success, national team versus club (player and season fixed effects, clustered by player). Panels (b)–(c): foreign replacement and defensive position shift on in-game substitution events (team and season fixed effects, clustered by team). Outcomes are in percentage points.

$^{**}\, p < .01$

FIGURE 5
Foreign–Domestic Talent Gap and Foreign Tenure Across Leagues



Notes: Panel (a) plots the median peak Transfermarkt market value of foreign and domestic outfield players by league (appearance-weighted), connected by a line, on a log scale; the multiplier is the foreign-to-domestic ratio of medians. Panel (b) plots the median number of days a foreign player has been at his current club. China (CSL, 2015–2025), Japan (J1, 2017–2025), and Korea (K1, 2020–2025); a descriptive comparison across the three leagues.

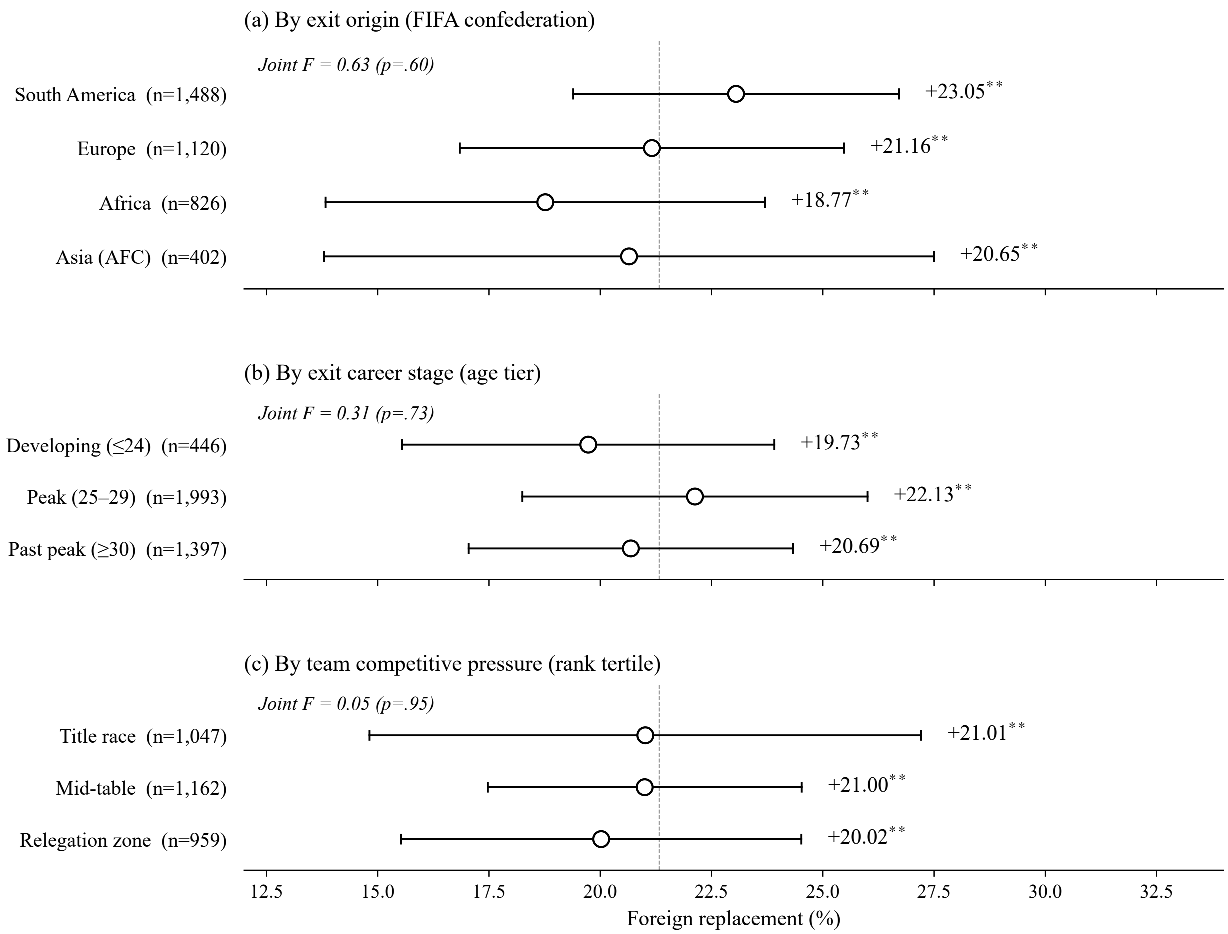
APPENDIX A

Substitution Across Player Subgroups

Figure A.1 tests whether foreign-for-foreign replacement varies by the exiting foreigner's origin, career stage, or the team's competitive pressure (the three moderators raised in review). It does not: the rate does not differ significantly across any subgroup (joint F, all $p > .40$), including for Asian (AFC) imports and across the standings, from title race to relegation zone.

FIGURE A.1

Substitution Heterogeneity by Foreign-Player and Team Characteristics (within CSL)



Notes: Foreign-out substitution events in China (CSL, 2015–2025), all outgoing foreigners. Panels disaggregate by FIFA confederation (including the Asian Football Confederation), age tier, and within-season tertile of the substituting team's pre-match league rank. Markers are subgroup means of foreign replacement (equal to 100 if the incoming replacement is foreign), with 95% cluster-robust confidence intervals (cluster: team). In each panel a joint F tests equality of the subgroup means, so a high p indicates that foreign replacement does not vary across that subgroup.

$^{**}\ p < .01$

The Talent Gap and Substitution

Table A.1 tests the team-season talent gap as a moderator of the substitution signature. In CSL, a one-standard-deviation wider gap amplifies the foreign-for-foreign effect by 4.91 percentage points ($p < .01$), while the defensive-shift placebo is unmoved; neither J1 nor K1 shows gap moderation.

TABLE A.1
Within-CSL Talent-Gap Dose-Response in the Foreign-Centered Architecture

	(1) Foreign replacement	(2) Defensive position shift
Panel A: China (CSL)		
Withdrawn player is foreign	14.76**	14.15**
	(1.27)	(1.47)
× Talent gap	4.91**	0.69
	(1.26)	(1.48)
Outcome mean	12.17	22.98
Observations	16,494	16,494
Panel B: Japan (J1)		
Withdrawn player is foreign	2.59	4.48*
	(2.11)	(1.91)
× Talent gap	-1.34	-2.08
	(0.82)	(1.43)
Outcome mean	18.23	19.61
Observations	23,471	23,471
Panel C: Korea (K1)		
Withdrawn player is foreign	6.60**	4.95†
	(2.02)	(2.48)
× Talent gap	0.75	-2.52
	(1.21)	(2.35)
Outcome mean	20.77	22.04
Observations	10,456	10,456

Notes: Sample is in-game substitution events with a non-missing team-season talent-gap proxy in China (CSL), Japan (J1), and Korea (K1). Outcome definitions and base controls match Table 4. *Withdrawn player is foreign* equals 1 if the outgoing player is foreign; *Talent gap* is the team-season natural log of the mean foreign-to-domestic peak Transfermarkt market value, z-scored within country, so the interaction reads as the change in the effect per within-country standard deviation of the gap. Column (1) outcome is foreign replacement; column (2) is defensive position shift. All specifications include team and season fixed effects; standard errors clustered by team.

$^{\dagger}\, p < .10$
$^{*}\, p < .05$
$^{**}\, p < .01$